MLLM SEGMENTATION · HALLOWEEN 자율세션 2026.10.30 · NO. 1/4

Better, Stronger, Faster, Broader: Structured All-Mask Prediction (STAMPlus)

Decouple dialogue from mask prediction — one non-autoregressive pass for every target

> **Central claim** — MLLM segmentation faces a trilemma: segmentation performance, preserved dialogue, fast inference. STAMP/STAMPlus reconcile all three by emitting one trigger token and predicting **all masks in a single non-autoregressive pass**.

Jiazhen Liu · Mingkuan Feng · Long Chen — `arXiv:2608.02791` · Aug 2026

81.0
RefCOCO-family avg cIoU · Qwen2-7B · Table II

64.0
ReasonSeg avg (4 scores) · Table IV

13.50 → 5.16 s
12-category latency · repeated STAMP vs one-pass · abstract

76.2
RRSIS-D avg · Table V

Trilemma accuracy ↔ dialogue ↔ latency › **STAMP** binary all-mask in one pass › **STAMPlus** structured multi-target · IDs + boxes

01 · PROBLEM The segmentation trilemma

Segmentation methods on MLLMs must be **accurate**, must not damage the LLM's **dialogue ability**, and must stay **fast**. The two dominant families each break one leg:

- **Embedding-prediction** — pixel-level objectives can disrupt language modeling.
- **Next-token generation of masks** — efficient dialogue, but inefficient for dense masks (autoregressive, repeated).

And binary all-mask prediction cannot store **multiple identities** without repeating a target-specific pass — the "repeated STAMP" cost on the right.

02 · METHOD A STAMP — Simultaneous Textual All-Mask Prediction

Image patch features › in-vocab <SEG> trigger › mask tokens fused w/ patches › hybrid attention: FG/BG in 1 pass

An **in-vocabulary trigger** keeps the autoregressive dialogue untouched. Image-aligned mask tokens are fused with corresponding patch features, and **hybrid attention** classifies all tokens as foreground/background simultaneously — non-autoregressive, single pass.

> **Why it survives the trilemma:** textual path stays a normal LM token; mask path is one cheap classification sweep.

03 · READING NOTES Deep-frame view

The move mirrors DETR-era set prediction: push per-instance loops into a single parallel prediction with explicit structure (IDs/boxes) — but here the loop lives *inside a frozen-capability MLLM*, so dialogue quality is the constraint being guarded.

04 · PARADIGM MAP Table I at a glance

Embedding-prediction family	Mask steps	Token-only supv.	Decoder-free
LISA · GSVA · PixelLM · M2SA · READ	$\mathcal{O}(N)$	×	×

All five current embedding-prediction models decode masks **outside the token interface** at $\mathcal{O}(N)$ steps per instruction. Token-prediction methods keep everything in the interface but pay step-by-step for dense masks. All-mask prediction (Fig. 2c) decouples: dialogue stays textual, masks come out in one sweep — **decoder-free**.

05 · METHOD B STAMPlus — structured multi-target

Binary masks cannot represent several semantic/instance identities. STAMPlus makes the mask space **structured** (Fig. 5):

target list + explicit IDs › optional boxes › IDs → shared multi-class space › all targets NAR pass

- **Open-vocabulary semantic** segmentation
- **Instance-aware** segmentation (ID binding)
- **Remote-sensing small-target** — high-resolution mask-token scaling preserves finer spatial evidence

06 · RESULTS One pass instead of twelve

5.16 s
12-category latency · single structured pass

13.50 s
repeated STAMP for 12 categories

−61.8%
relative latency reduction (5.16/13.50 − 1)

Fig. 9: STAMP and LISA repeat a **similar-cost prediction per category**, so latency climbs steeply with the number of targets — STAMPlus's added cost is **marginal**. A single unified checkpoint retains STAMP's referring and reasoning capabilities.

Abstract also reports state-of-the-art segmentation across the three structured settings while preserving general multimodal instruction following.

07 · ANALYSES Target cues & look-twice

Table VIII(a): swapping the Phase-1 target description for a **human-provided box improves** referring segmentation; Table VIII(b): segmentation-derived spatial grounding **supports reasoning** — "look-twice" benefits from learned grounding.

> **Boundary:** Table X ablation timing measured on a **single NVIDIA A800**; Fig. 8 comparisons depend on input-resolution and model scale. Latency gains are structural, not a universal wall-clock promise.

08 · BENCHMARKS Three settings, one checkpoint

RefCOCO-family Avg cIoU · Table II	8-split avg
LISA · Vicuna-7B	69.9
Text4Seg · Vicuna-7B	74.9
Text4Seg++ · Qwen2-7B	78.9
STAMP · Qwen2-2B	79.1
STAMPlus · Qwen2-2B	79.5
STAMP · Qwen2-7B	80.7
STAMPlus · Qwen2-7B	**81.0**

Same checkpoint elsewhere — ReasonSeg (Table IV, avg of val/test gIoU·cIoU): READ 61.1 → **STAMPlus-7B 64.0**. RRSIS-D (Table V, avg of 6 metrics): specialized RMSIN 72.4 → **STAMPlus-7B 76.2**.

09 · ABLATIONS Table X — every part pays

STAMP ablation (RefCOCO avg)	cIoU	Time (s)
full, 2B @ 896²	**79.1**	1.3
w/o Ahyb	76.2	1.3
w/o Emask	73.3	1.3
w/o SAM	75.3	0.9
7B @ 896²	**80.7**	2.4

Removing the mask embedding costs the most (**−5.8 pt**); removing SAM drops −3.8 pt while cutting 0.4 s of decode — the two design axes trade cost and quality in opposite directions.

10 · SCALING Table XI — same checkpoint, richer tokens

Mask-token budget	RRSIS-D cIoU	EarthReason cIoU
1024–1280 (standard)	76.5	70.7
2560–3200 (high-res)	**81.5**	**71.2**

Scaling input resolution + mask-token budget with the **same STAMPlus-2B checkpoint** lifts RRSIS-D by **+5.0 cIoU** (gIoU 58.1→63.6, Acc@0.5 63.2→71.0) and EarthReason +0.5 — spatial resolution, not heavier training, is the lever (Table XI).

11 · HOW TO READ Averages are protocol-dependent

81.0 = mean cIoU over eight RefCOCO-family splits · **64.0** = mean of 4 ReasonSeg scores (val/test × gIoU/cIoU) · **76.2** = mean over six RRSIS-D val/test metrics. Baselines come from the paper's own tables, not re-runs.

12 · OPEN LIMITS What remains unresolved?

Ablation credits per module (Table X) and the high-resolution mask-token budget trade-off (Table XI) leave engineering choices open; absolute latency depends on GPU/runtime, and the paper's latency story is category-count sensitive.

13 · SESSION Team poster #1 of 4

할로윈 자율세션 · 팀 공유 논문 포스터. 같은 세션: Search-R1 · Perception Encoder · Left-Right Symmetry Breaking. 아무것도 안 무섭다 — 원문 수치만.

Reference

Jiazhen Liu, Mingkuan Feng, Long Chen. "Better, Stronger, Faster, and Broader: Structured All-Mask Prediction for MLLM-Based Segmentation." arXiv:2608.02791 (Aug 2026). 모든 수치는 원문 Table II·IV·V·VIII·Table X·XI + abstract에서 전사·재확인 (2026-10-30 자율세션 · ID 검증 원칙).